

Furunculosis—Etiology and Treatment

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EVIDENCE indicates that there are many factors in the etiology of furunculosis. There seems to be a widespread belief among laymen that boils are the result of 'bad blood,' but no scientific support has been presented to substantiate such a view. Cleanliness of the skin was considered a major factor by Gant, Owens and Schwartz (1) in a severe outbreak of boils among a group of tunnel workers. The resistance of the host is an important factor in any infection and it seems logical to assume that low tissue resistance would play a rôle in the development of furunculosis. The present report reviews 16 cases in which tissue resistance seemed to be an important factor.

METHODS

The 16 patients were college students between the ages of 17 and 25 years. They represented the only cases of furunculosis seen among approximately 2000 students over a period of one year. A careful history was taken and a physical examination made; particular attention was given to evidence of dietary deficiency. Hemoglobin was estimated by the Newcomber method. The basal metabolism was determined using the usual precautions and the new Sanborn waterless apparatus. Basal temperature was obtained by the method previously described in which the thermometer is left under the tongue for 10 minutes before the patient gets out of bed in the morning. Oral and rectal temperatures were taken at the time of examination.

RESULTS

No correlation was apparent between the hemoglobin values and the development of boils. The average was 14.0 gm. of hemoglobin per 100 cc., the range 11.5 to 17.0 gm. Many

students free from boils have been seen in whom the hemoglobin values were in the lower part of this range. The largest boil observed in this group of patients was in the axilla of a boy whose hemoglobin was 15.3 gm.; the basal metabolic rate in this case was minus 24.

TABLE 1. BASAL TEMPERATURE AND B.M.R. IN
16 CASES OF FURUNCULOSIS

Patient	Temperature		Sex
	°F.	B.M.R.	
MH	97.0	-12	M
JH	97.0	-24	M
DC	97.2	-19	M
BS	97.0	-20	M
CT	97.0	-19	M
MJ	97.4	-13	M
AG	97.5	-10	M
BS	97.0	-5	M
DC	97.5	-9	M
WS	97.5	+7	M
GJ	97.0	-20	M
JO	97.0	-15	M
DD	97.4	-9	F
GM	97.0	-15	F
AG	97.4		F
GM	97.2		F

The history and physical examination disclosed no evidence of malnutrition in this group of patients. There were no obvious local skin conditions contributing to the development of furunculosis. One factor was common in the physical examinations, namely, subnormal temperatures (table 1). Temperatures taken at the time of physical examination were unreliable since excitement or exercise will elevate the temperature. However, in most people with a subnormal temperature the rectal temperature is below 99.0° F. at the time of examination. Basal temperatures were uniformly low on this group of cases. The normal range for basal temperature is thought to be about 97.8 to 98.2° F.

The determinations of the basal metabolic rate were low, with one exception in which the reading was plus 7. However, in this case the body temperature was low and the high basal

metabolic rate was thought to result from nervousness and inability to relax. Most of the basal values were between minus 24 and minus 10 on the first test (table 1). It is thought that all cases would have been in this range if repeated determinations had been obtained. However, basal temperature is now being used as a criterion of lowered tissue resistance more than basal metabolism.

Therapy. The furuncles were given local treatment of heat and incision for drainage after pus was apparent. A few healed by resorption without being opened. Heat therapy consisted of hot wet dressings at home and 30 minutes to one hour under a Zeolite infra-red lamp at some time during the day at the Health Department.

Two cases served as controls without systemic therapy for 2 to 3 months. Boils recurred during this period; heat therapy was given as often as the time of these patients would permit but in each case the furuncle would enlarge and require incision. In one of the control cases thyroid therapy (1 grain a day) was started and heat therapy stopped when there were two tender lumps under one arm and one under the other. These were about half the size usually seen in this patient at the time of incision. Within 10 days the three tender lumps had completely disappeared, and no evidence of a recurrence has been observed. In the other control case thyroid therapy was started during a period of remission and no further furuncles appeared.

In the remaining 14 cases of furunculosis thyroid therapy was started when the patient was first seen. A second boil did not develop in any case during the period of thyroid therapy. One student who had repeated boils before therapy was started had no recurrences during 8 months of treatment. He then stopped medication without medical advice and boils recurred within two months. Details of the results with continued therapy have been published (2).

DISCUSSION

In the study of furunculosis in the present series it would seem that most of the usual etiologic factors, including cleanliness of the

skin, can be excluded. The low metabolic rate and subnormal temperatures indicate the tissue resistance might be low. Experimental it has been shown that skin disorders are numerous in the cretin rabbit (3). In cases of human myxedema the blood flow per minute and the skin temperature are markedly reduced and are promptly restored to normal during thyroid medication (4). Since the healing process is accompanied by an increase in circulation, it is not surprising that thyroid therapy would aid in skin infections in cases with poor peripheral circulation. Furthermore, it seems probable that an increased flow of blood through the skin would aid in the prevention of localized infection. The prompt response to thyroid therapy of the furuncles in this series of cases seems to furnish evidence for such a theory. The resistance of the skin to local infection seems to have improved since no more furuncles developed during the period of thyroid therapy. It is impossible to state whether the resistance was due to an action on the individual cells, to an increased blood flow, or to the combination of the two. Ball (5) reported that 'boils cleared with dramatic suddenness' using thyroid and potassium permanganate. It is impossible to state how much of the success with this therapy was due to both constituents but the same results have been obtained in the cases reported here with the use of thyroid alone.

SUMMARY

Furunculosis was observed in 16 patients. The basal metabolic rate or basal temperature was below normal in each case. Further lesions did not develop after starting thyroid therapy unless the medication was discontinued. It would appear that skin resistance may be a factor in furunculosis if the peripheral blood supply is below normal.

REFERENCES

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